

ASSIGNMENT 5

Automation Testing Fundamentals

Q1. Explain the difference between manual testing and automation testing. What are the main advantages of automation testing?

Manual testing is a software testing approach in which a tester manually executes test cases without using any automation tools. The tester interacts with the application and verifies whether it works as expected.

Automation testing is a software testing approach in which test cases are executed automatically using testing tools and scripts such as Selenium. The tool performs the testing tasks and compares the actual results with the expected results.

The main advantages of automation testing are:

1. It executes test cases faster than manual testing.
2. It reduces human errors and improves accuracy.
3. It saves time and effort for repetitive test cases.
4. It supports repeated execution of tests, making regression testing easier.
5. It allows tests to run on multiple browsers and environments.

Q2. Describe the steps involved in setting up a Selenium test environment (installation, configuration, and browser setup).

The steps involved in setting up a Selenium test environment are:

1. Install the Java Development Kit (JDK).
2. Install an IDE such as Eclipse.
3. Create a Java project in the IDE.
4. Add Selenium WebDriver libraries to the project using Maven or JAR files.
5. Download and install a web browser such as Google Chrome.
6. Download the appropriate browser driver (ChromeDriver for Chrome).
7. Configure the browser driver in the Selenium project.
8. Write and execute a sample Selenium script to verify the setup.

Q3. Write a simple Selenium WebDriver script in Java to open Google, search for “Automation Testing,” and close the browser.

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class GoogleSearch {
    public static void main(String[] args) {
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");
        WebElement searchBox = driver.findElement(By.name("q"));
        searchBox.sendKeys("Automation Testing");
        searchBox.submit();
        driver.quit();
    }
}
```

Q4. What are the main components of Selenium? Explain the purpose of Selenium Grid with an example use case.

The main components of Selenium are:

- Selenium IDE – Used for recording and playback of test cases.
- Selenium WebDriver – Used for automating web browser actions.

- Selenium Grid – Used for running tests on multiple machines and browsers simultaneously.

Purpose of Selenium Grid:

Selenium Grid allows testers to execute tests in parallel across different browsers, operating systems, and machines. This reduces execution time and improves test coverage.

Example Use Case:

A company wants to test its website on Chrome, Firefox, and Edge browsers. Using Selenium Grid, the same test can run simultaneously on all three browsers, saving time and ensuring compatibility.

Q5. Using Selenium IDE, record a login test case for any demo website. List the key steps and explain how “Record and Play” works in Selenium IDE.

Key Steps:

1. Install the Selenium IDE browser extension.
2. Open Selenium IDE and create a new project.
3. Click the “Record” button.
4. Open a demo website login page.
5. Enter the username and password.
6. Click the Login button.
7. Stop the recording and save the test case.
8. Click “Run” to replay the recorded test.

Record and Play Works:

The “Record” feature captures all user actions such as clicks, typing, and navigation. Selenium IDE automatically converts these actions into test commands. The “Play” feature re-executes the recorded commands in the

same sequence, allowing automated verification of the login process without manually repeating the steps.